

SAP ABAP PROJECT

Step to create: A Smartform with Multiple Windows

This project guide shows you how to create a complete SAP Smartform from scratch. You will learn how to design a layout with separate windows for a title bar, an employee photo, a barcode, fixed template fields, and a dynamic table.

Phase 1: Create a Custom Style (T-code: SMARTSTYLES)

Before designing the form layout, we need to define the fonts and text sizes. We do this in the Smart Styles transaction.

Step to create 1: Open T-code SMARTSTYLES

Type SMARTSTYLES in the SAP Command Field at the top of your screen and press Enter.

Step to create 2: Enter Style Name and Create

In the **Style Name** field, type ZEMP_STYLE. Then, click the **Create** button.

SAP GUI - Maintain Smart Styles: Initial Screen

```
Style Name      [ ZEMP_STYLE
```

```
[ [Create] ]      [ Change ]      [ Display ]
```

1. Right-click on the **Paragraph Formats** folder on the left side and choose **Create Node**.
2. Create a tag named T1 for the Title: Set Font Family to *Times New Roman*, Font Size to *16 pt*, and check the **Bold** box.
3. Create another tag named B1 for Regular Body Text: Set Font Family to *Times New Roman*, Font Size to *10 pt*, and leave it regular.
4. Double-click the **Header Data** folder at the top of the tree. In the **Standard Paragraph** field, type B1. This makes B1 your default text style.

Click the **Save** icon (or press Ctrl + S), then click the **Activate** icon (or press Ctrl + F3). Your style is now ready to use.

Type SMARTFORMS in the SAP Command Field and press Enter.

In the **Form Name** field, type ZEA_EMPLOYEE_FORM. Then, click the **Create** button.

SAP GUI - Smartforms: Initial Screen

Form Name

[ZEA_EMPLOYEE_FORM]

[[Create]]

[Change]

[Display]

Step to create 7: Assign your Custom Style to the Form

1. Expand the **Global Settings** folder on the left and double-click **Form Attributes**.
2. Go to the **Output Options** tab.
3. Find the **Style** field and type ZEMP_STYLE (the style we made in Phase 1).

Step to create 8: Set Up Form Interface Import Parameters

Double-click the **Form Interface** node on the left. Under the **Import** tab, add the fields you want to pass from your ABAP program:

- IM_EMP_ID TYPE PERNR_D (Employee ID)
- IM_EMP_NAME TYPE STRING (Employee Name)

Tip: Putting different items into their own windows stops text from overlapping and keeps your layout clean.

Phase 3: Create Layout Windows, Templates, and Tables

Now we will create separate layout sections on our page using the Form Painter workspace.

Step to create 9: Create the Title Bar Window

1. Right-click on **Page 1** → **Create** → **Window**. Name this window WIN_TITLE.
2. Set its position and size fields: Width: 18.00 cm, Height: 1.50 cm, Top: 1.00 cm, Left: 1.50 cm.
3. Right-click on your new WIN_TITLE window node → **Create** → **Text**.
4. In the text editor box, change the paragraph tag dropdown to T1 and type: EMPLOYEE DETAILS MASTER REPORT.

Step to create 10: Create the Employee Photo Window

1. Right-click on **Page 1** → **Create** → **Window**. Name it WIN_PHOTO.
2. Position it on the top-right corner: Width: 3.50 cm, Height: 4.00 cm, Top: 3.00 cm, Left: 16.00 cm.
3. Right-click on WIN_PHOTO → **Create** → **Graphic**.
4. In the properties tab, enter your image details:
 - **Name:** ZEMP_PHOTO MOCK (The name of your uploaded bitmap file)
 - **Object:** GRAPHICS | **ID:** BMAP | **Type:** BCOL (Color)

Step to create 11: Create the Barcode Window

1. Right-click on **Page 1** → **Create** → **Window**. Name it WIN_BARCODE.
2. Position it under the photo window: Width: 3.50 cm, Height: 2.00 cm, Top: 7.50 cm, Left: 16.00 cm.
3. Right-click on WIN_BARCODE → **Create** → **Text**.
4. Type your variable name exactly like this: &IM_EMP_ID&.
5. Highlight the text and choose a standard system barcode format font, like BC_CD39, from the character style dropdown.

Step to create 12: Create a Template Window (For Fixed Data Alignment)

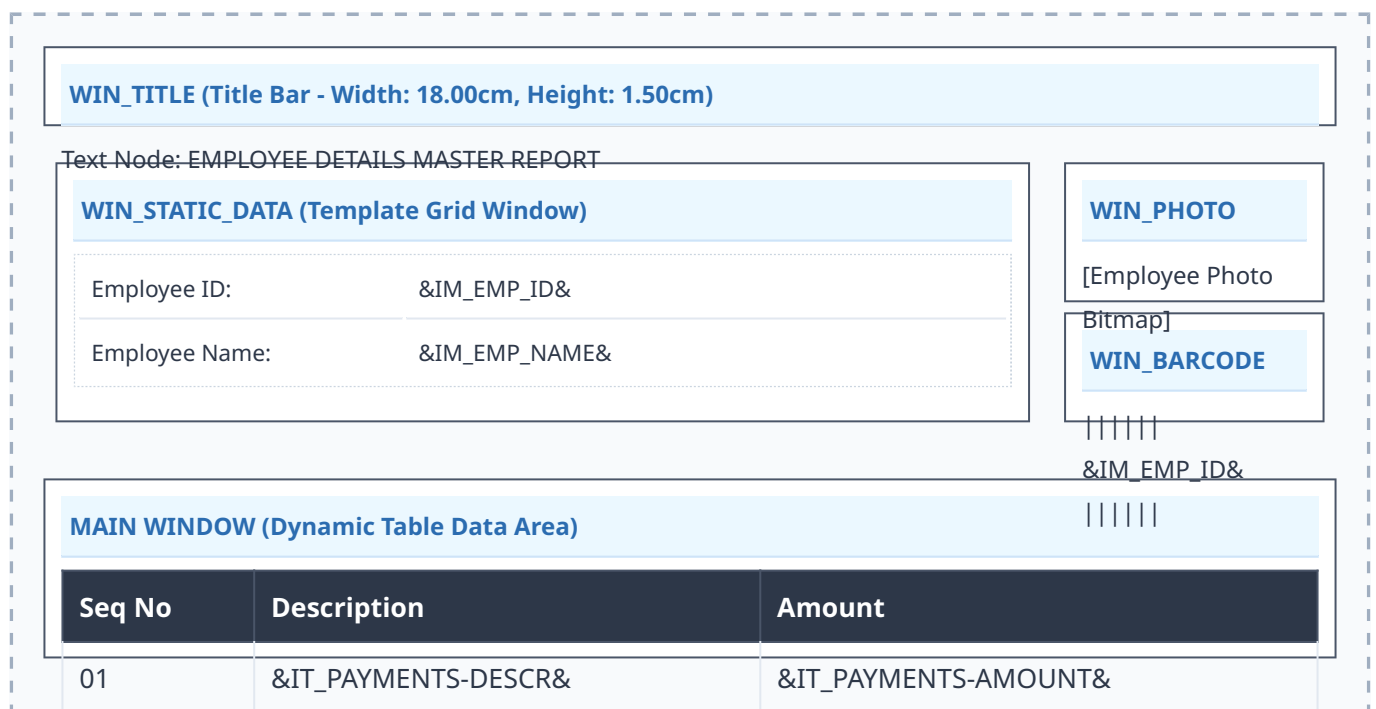
1. Right-click on **Page 1** → **Create** → **Window**. Name it WIN_STATIC_DATA.
2. Set the size: Width: 13.50 cm, Height: 5.00 cm, Top: 3.00 cm, Left: 1.50 cm.
3. Right-click on WIN_STATIC_DATA → **Create** → **Template**.
4. Go to the **Details** tab and set up your grid rows and columns: define 2 columns with widths of 4.50 cm and 9.00 cm.
5. Create child text nodes inside the template grid cells to match your fields:
 - Row 1, Column 1 Text: Employee ID:
 - Row 1, Column 2 Text: &IM_EMP_ID&
 - Row 2, Column 1 Text: Employee Name:
 - Row 2, Column 2 Text: &IM_EMP_NAME&

Step to create 13: Set Up a Dynamic Data Table in the MAIN Window

1. Click on your default MAIN window. Move and resize it to sit at the bottom section: Width: 18.00 cm, Height: 14.00 cm, Top: 10.00 cm, Left: 1.50 cm.
2. Right-click on MAIN → **Create** → **Table**. Name it TBL_PAYMENT_ITEMS.
3. In the **Table** tab, draw Line Type L1 with 3 columns measuring: 3.00 cm, 10.00 cm, and 5.00 cm.
4. Expand the table node to see its three standard structural folders:
 - **Header:** Right-click and insert fixed text titles for your columns (e.g., *Seq No, Description, Amount*).
 - **Main Area:** Right-click and create a Table Line node using line type L1. Inside the individual cells, add text nodes for your loop variables: &IT_PAYMENTS-LINE_ITEM&, &IT_PAYMENTS-DESCR&, and &IT_PAYMENTS-AMOUNT&.
 - **Footer:** Use this section if you need to print a grand total line.

Form Painter Layout Blueprint View

Below is a visual layout map showing how your different windows sit on the page:



Phase 4: Activate and Test via Driver Program

Step to create 14: Check and Activate your Form

Press Ctrl + F2 to check for configuration or structural errors. If everything looks good, press Ctrl + F3 to activate the Smartform. SAP will automatically build a unique background Function Module for it.

Step to create 15: Create the ABAP Driver Program

Go to transaction code SE38, create a new executable program named Z_CALL_EMPLOYEE_SMARTFORM, and paste the code below to test your new form layout:

```
REPORT z_call_employee_smartform.

DATA: lv_fm_name TYPE rs38l_fnam.

" Find the exact function module name generated by your Smartform
CALL FUNCTION 'SSF_FUNCTION_MODULE_NAME'
  EXPORTING
    formname      = 'ZEA_EMPLOYEE_FORM'
  IMPORTING
    fm_name       = lv_fm_name
  EXCEPTIONS
    no_form       = 1
    no_function_module = 2
    OTHERS        = 3.

IF sy-subrc = 0.
  " Run the generated function module to open the print preview screen
  CALL FUNCTION lv_fm_name
    EXPORTING
      im_emp_id   = 'E10024'
      im_emp_name = 'Sanket Singh'
    EXCEPTIONS
      formatting_error = 1
      internal_error   = 2
      send_error       = 3
      user_canceled    = 4
      OTHERS           = 5.
ENDIF.
```